

SWT 22022 – Practical for
Internet Application
Development

Department of Information
and Communication
Technology
Faculty of Technology



Lab sheet :20
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Title:

- E – Commerce App (React) - I

Aims:

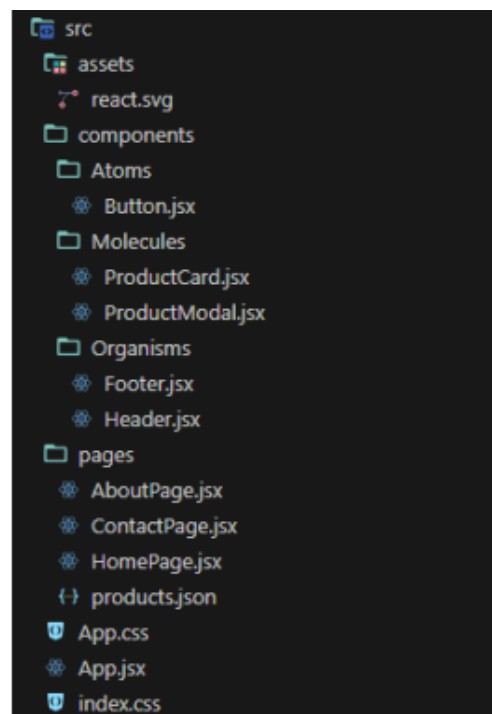
- Implement a ReactJS frontend application

Tasks:

- Create a New Project
- Install Necessary Packages
- Implement Components
- Integrate Components into the Page

Activities :

1. Create new project named “e-commerce-app”
2. Install Bootstrap
npm install bootstrap
npm install react-bootstrap bootstrap
3. Install react-route-dom
npm install react-router-dom
4. Create folder structure as follows



5. Implement Footer.jsx

```
import { Navbar, Container } from "react-bootstrap";
const Footer = () => {
  return (
    <Navbar bg="dark" variant="dark" fixed="bottom">
      <Container>
        <p className="text-white">Copyright &copy; 2024 My E-commerce
App</p>
      </Container>
    </Navbar>
  );
};
export default Footer;
```

6. Implement Header.jsx

```
import { Navbar, Container, Nav } from 'react-bootstrap';
import { Link } from 'react-router-dom';
const Header = () => {
  return (
    <Navbar bg="dark" variant="dark" fixed="top">
      <Container>
        <Navbar.Brand href="#home">My E-commerce App -Admin</Navbar.Brand>
        <Nav>
          <Nav.Link as={Link} to="/">Products</Nav.Link>
          <Nav.Link as={Link} to="/about">About</Nav.Link>
          <Nav.Link as={Link} to="/contact">Contact</Nav.Link>
        </Nav>
      </Container>
    </Navbar>
  );
};
export default Header;
```

7. Implement ProductCard.jsx

```
import Card from 'react-bootstrap/Card';
const ProductCard = ({ product }) => {
  return (
    <Card>
      <Card.Img variant="top" src={product.image} />
      <Card.Body>
        <Card.Title>{product.name}</Card.Title>
        <Card.Text>{product.description}</Card.Text>
        <Card.Text>${product.price}</Card.Text>
      </Card.Body>
    </Card>
  );
};
```

```

        </Card>

    );

};

export default ProductCard;

```

8. Implement ProductModal.jsx

```

import { useState } from "react";
import { Button, Modal, Form } from "react-bootstrap";

const ProductModal = ({ show, onHide, onSubmit, initialProduct }) => {
  const [formData, setFormData] = useState(
    initialProduct || {
      name: "",
      description: "",
      price: "",
    }
  );

  const handleChange = (e) => {
    setFormData({ ...formData, [e.target.name]: e.target.value });
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    onSubmit(formData);
    onHide();
  };

  return (
    <Modal show={show} onHide={onHide}>
      <Modal.Header closeButton>

```

```

    <Modal.Title>initialProduct ? "Edit Product" : "Add Product"</Modal.Title>

  </Modal.Header>

  <Modal.Body>

    <Form onSubmit={handleSubmit}>

      <Form.Group controlId="formProductName">

        <Form.Label>Product Name</Form.Label>

        <Form.Control type="text" name="name" placeholder="Enter product name"
value={formData.name} onChange={handleChange} required />

      </Form.Group>

      <Form.Group controlId="formProductDescription">

        <Form.Label>Product Description</Form.Label>

        <Form.Control as="textarea" name="description" placeholder="Enter product description"
value={formData.description} onChange={handleChange} />

      </Form.Group>

      <Form.Group controlId="formProductPrice">

        <Form.Label>Product Price</Form.Label>

        <Form.Control type="number" name="price" value={formData.price} onChange={handleChange}
required/>

      </Form.Group>

    </Form>

  </Modal.Body>

  <Modal.Footer>

    <Button variant="secondary" onClick={onHide}>Close</Button>

    <Button variant="primary" onClick={handleSubmit}>{initialProduct ? "Save Changes" : "Add
Product"}</Button>

  </Modal.Footer>

</Modal>

);

};

export default ProductModal;

```

9. Implement AboutPage.jsx

```

import Header from '../components/Organisms/Header';
import Footer from '../components/Organisms/Footer';
const AboutPage = () => {
  return (
    <>
      <Header />
      <div className="mx-5">
        <h1>About Us</h1>
        <p>Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim a minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. </p>
        <p><strong>Our Mission:</strong> To provide high-quality products and exceptional customer service.</p>
        <p><strong>Our Vision:</strong> To be the leading e-commerce platform in the industry.</p>
      </div>
      <Footer />
    </>
  );
};

export default AboutPage;

```

10. Implement ContactPage.jsx

11. Implement HomePage.jsx

```

import Header from "../components/Organisms/Header";
import Footer from "../components/Organisms/Footer";
import ProductCard from "../components/Molecules/ProductCard";
import productsData from "../products.json";
import { useState } from "react";
import { Button } from "react-bootstrap";
import ProductModal from "../components/Molecules/ProductModal";
const HomePage = () => {

```

```

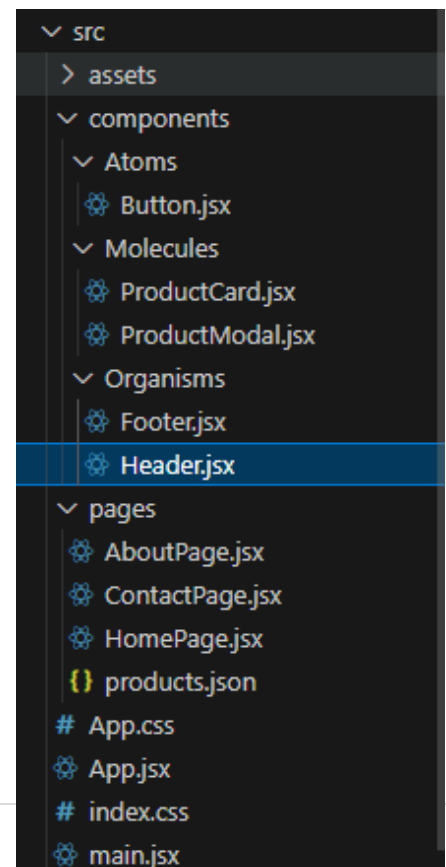
const [showProductModal, setShowProductModal] = useState(false);
const [currentProduct, setCurrentProduct] = useState(null);
const handleAddProduct = () => {
    setCurrentProduct(null);
    setShowProductModal(true);
};
const handleEditProduct = (product) => {
    setCurrentProduct(product);
    setShowProductModal(true);
};
const handleProductSubmit = (productData) => {
    setShowProductModal(false);
};
return (
    <>
        <Header />
        <div className="mt-5">
            <Button onClick={handleAddProduct}>Add Product</Button>
            <ProductModal show={showProductModal} onHide={() =>
setShowProductModal(false)} onSubmit={handleProductSubmit} initialProduct={currentProduct} />
            <h1 className="text-center mt-4">All Products</h1>
            <div className="row mb-3">{productsData.map((product) => (
                <div className="col-md-3 mb-3" key={product.id}>
                    <ProductCard product={product} />
                    <Button onClick={() =>
handleEditProduct(product)} className="mt-2">Edit Product</Button>
                </div>
            ))}</div>
        </div>
        <Footer />
    </>
);

```

```
};  
export default HomePage;
```

12. Implement App.jsx

```
import { BrowserRouter as Router, Routes, Route } from 'react-routerdom';  
import HomePage from './pages/HomePage';  
import AboutPage from './pages/AboutPage';  
import ContactPage from './pages/ContactPage';  
import './App.css';  
const App = () => {  
  return (  
    <Router>  
      <Routes>  
        <Route path="/" element={<HomePage />} />  
        <Route path="/about" element={<AboutPage />} />  
        <Route path="/contact" element={<ContactPage />} />  
      </Routes>  
    </Router>  
  );  
};  
export default App;
```



ProductCard.jsx

```
/* eslint-disable react/prop-types */
import Card from 'react-bootstrap/Card';
// import Button from 'react-bootstrap/Button';
const ProductCard = ({ product }) => {
  return (
    <Card>
      <Card.Img variant="top" src={product.image} />
      <Card.Body>
        <Card.Title>{product.name}</Card.Title>
        <Card.Text>{product.description}</Card.Text>
        <Card.Text>${product.price}</Card.Text>
        {/* <Button variant="primary">Add to Cart</Button> */}
      </Card.Body>
    </Card>
  );
};

export default ProductCard;
```

ProductModal.jsx

```
/* eslint-disable react/prop-types */
import { useState } from "react";
import { Button, Modal, Form } from "react-bootstrap";

const ProductModal = ({ show, onHide, onSubmit, initialProduct }) => {
  const [formData, setFormData] = useState(
    initialProduct || {
      name: "",
      description: "",
      price: "",
      // ... other fields
    }
  );

  const handleChange = (e) => {
    setFormData({ ...formData, [e.target.name]: e.target.value });
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    onSubmit(formData);
    onHide();
  };
};
```

```

return (
  <Modal show={show} onHide={onHide}>
    <Modal.Header closeButton>
      <Modal.Title>
        {initialProduct ? "Edit Product" : "Add Product"}
      </Modal.Title>
    </Modal.Header>
    <Modal.Body>
      <Form onSubmit={handleSubmit}>
        <Form.Group controlId="formProductName">
          <Form.Label>Product Name</Form.Label>
          <Form.Control
            type="text"
            placeholder="Enter product name"
            value={formData.name}
            onChange={handleChange}
            required
          />
        </Form.Group>
        {/* Add more form fields for description, price, etc. */}
      </Form>
    </Modal.Body>
    <Modal.Footer>
      <Button variant="secondary" onClick={onHide}>
        Close
      </Button>
      <Button variant="primary" type="submit">
        {initialProduct ? "Save Changes" : "Add Product"}
      </Button>
    </Modal.Footer>
  </Modal>
);
};

export default ProductModal;

```

Footer.jsx

```

import { Navbar, Container } from "react-bootstrap";

const Footer = () => {
  return (
    <Navbar bg="dark" variant="dark" fixed="bottom">
      <Container>
        <p className="text-white">Copyright &copy; 2024 My E-commerce App</p>
      </Container>
    </Navbar>
  );
};

```

```

    );
  };

  export default Footer;

```

Header.jsx

```

import { Navbar, Container, Nav } from 'react-bootstrap';
import { Link } from 'react-router-dom';

const Header = () => {
  return (
    <Navbar bg="dark" variant="dark" fixed='top'>
      <Container>
        <Navbar.Brand href="#home">My E-commerce App - Admin</Navbar.Brand>
        <Nav>
          <Nav.Link as={Link} to="/">Products</Nav.Link>
          <Nav.Link as={Link} to="/about">About</Nav.Link>
          <Nav.Link as={Link} to="/contact">Contact</Nav.Link>
        </Nav>
      </Container>
    </Navbar>
  );
};

export default Header;

```

AboutPage.jsx

```

import Header from '../components/Organisms/Header';
import Footer from '../components/Organisms/Footer';

const AboutPage = () => {
  return (
    <>
      <Header />
      <div className="mx-5">
        <h1>About Us</h1>
        <p>
          Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do
          eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad
          minim veniam, quis nostrud exercitation ullamco laboris nisi ut
          aliquip ex ea commodo consequat. Duis aute irure dolor in
          reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
          pariatur. Excepteur sint occaecat cupidatat non proident, sunt in
          culpa qui officia deserunt mollit anim id est laborum.
        </p>
      </div>
    </>
  );
};

```

```

    <p>
      <strong>Our Mission:</strong> To provide high-quality products and
      exceptional customer service.
    </p>
    <p>
      <strong>Our Vision:</strong> To be the leading e-commerce platform in
      the industry.
    </p>
  </div>
  <Footer />
</>
);
};

export default AboutPage;

```

ContactPage.jsx

```

import Header from "../components/Organisms/Header";
import Footer from "../components/Organisms/Footer";

const ContactPage = () => {
  return (
    <>
      <Header />
      <div className="mx-5">
        <h1>Contact Us</h1>
        <p>
          <strong>Have a question or need assistance?</strong> Please feel free
          to contact us using the information below.
        </p>

        <p>
          <strong>Address:</strong>
          Your Company Address
        </p>
        <p>
          <strong>Phone Number:</strong>
          (123) 456-7890
        </p>
        <p>
          <strong>Email:</strong>
          your_email@example.com
        </p>
      </div>
      <Footer />
    </>
  );
};

```

```
};  
  
export default ContactPage;
```

HomePage.jsx

```
import Header from "../components/Organisms/Header";  
import Footer from "../components/Organisms/Footer";  
import ProductCard from "../components/Molecules/ProductCard";  
import productsData from "../products.json";  
import { useState } from "react";  
import { Button } from "react-bootstrap";  
import ProductModal from "../components/Molecules/ProductModal";  
  
const HomePage = () => {  
  const [showProductModal, setShowProductModal] = useState(false);  
  const [currentProduct, setCurrentProduct] = useState(null);  
  
  const handleAddProduct = () => {  
    setCurrentProduct(null);  
    setShowProductModal(true);  
  };  
  
  const handleEditProduct = (product) => {  
    setCurrentProduct(product);  
    setShowProductModal(true);  
  };  
  
  const handleProductSubmit = (productData) => {  
    // Handle the submitted product data (e.g., send it to the backend)  
    setShowProductModal(false);  
    // Update the product list or perform other actions  
  };  
  
  return (  
    <>  
      <Header />  
      <div className="mt-5">  
        <Button onClick={handleAddProduct}>Add Product</Button>  
        <ProductModal  
          show={showProductModal}  
          onHide={() => setShowProductModal(false)}  
          onSubmit={handleProductSubmit}  
          initialProduct={currentProduct}  
        />  
        <h1 className="text-center mt-4">All Products</h1>  
        <div className="row mb-3">
```

```

        {productsData.map((product) => (
            <div className="col-md-3 mb-3" key={product.id}>
                <ProductCard product={product} />
                <Button onClick={() => handleEditProduct(product)} className="mt-
2">
                    Edit Product
                </Button>
            </div>
        ))}
    </div>
</div>
<Footer />
</>
);
};

export default HomePage;

```

products.json

```

[
  {
    "id": 1,
    "name": "Product 1",
    "description": "This is the first product.",
    "image": "/card.png",
    "price": 19.99
  },
  {
    "id": 2,
    "name": "Product 2",
    "description": "This is the second product.",
    "image": "/card.png",
    "price": 29.99
  },
  {
    "id": 4,
    "name": "Product 4",
    "description": "This is the third product.",
    "image": "/card.png",
    "price": 39.99
  },
  {
    "id": 5,
    "name": "Product 5",
    "description": "This is the fourth product.",
    "image": "/card.png",
    "price": 39.99
  }
]

```

```

    },
    {
      "id": 6,
      "name": "Product 6",
      "description": "This is the fifth product.",
      "image": "/card.png",
      "price": 39.99
    }
  ]

```

App.jsx

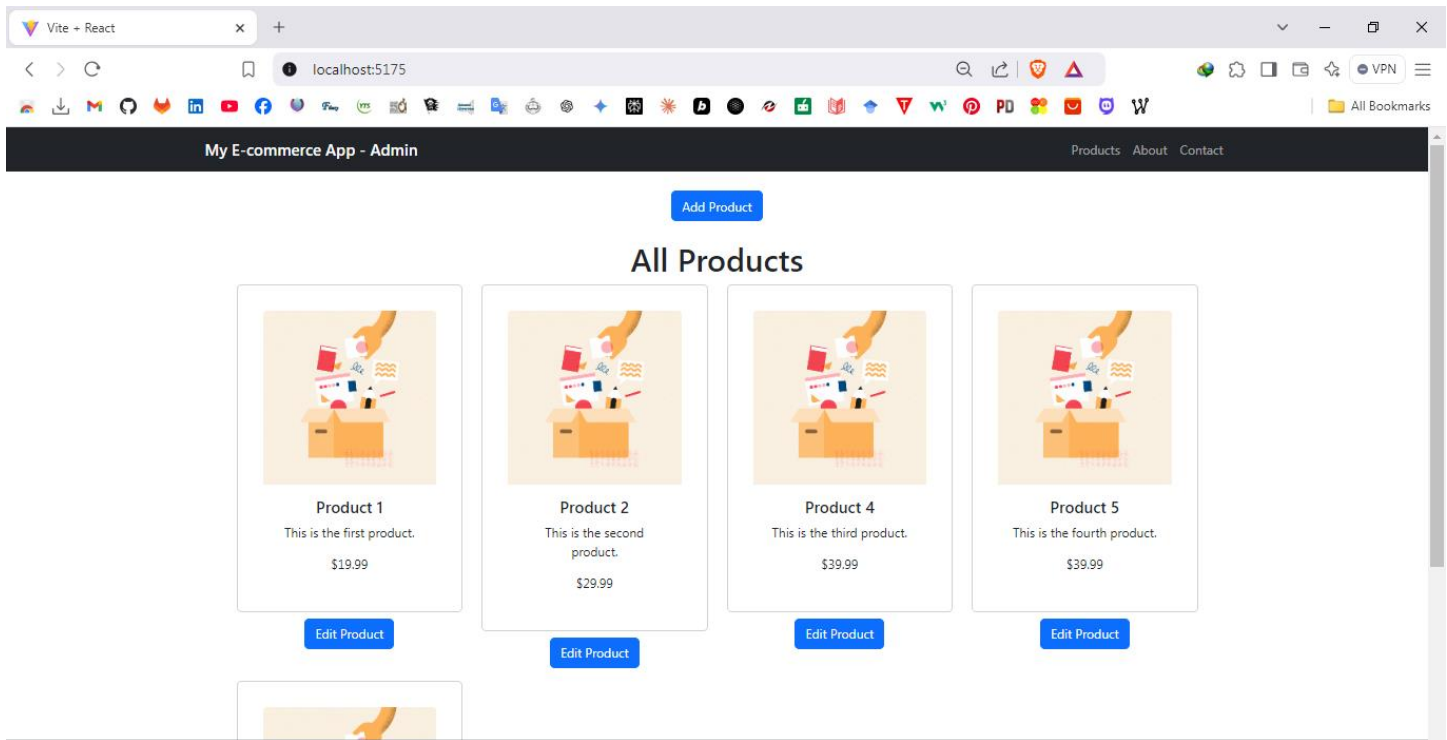
```

import { BrowserRouter as Router, Routes, Route } from 'react-router-dom';
import HomePage from './pages/HomePage';
import AboutPage from './pages/AboutPage';
import ContactPage from './pages/ContactPage';
import './App.css';

const App = () => {
  return (
    <Router>
      <Routes>
        <Route path="/" element={<HomePage />} />
        <Route path="/about" element={<AboutPage />} />
        <Route path="/contact" element={<ContactPage />} />
      </Routes>
    </Router>
  );
};

export default App;

```



Discussion:

- In this lab practical session, I gained hands-on experience in building dynamic applications with React. The session's main objective was to create a dynamic list using `useState`, which helped me manage the state of items in the list efficiently. I also explored the power of the Context API, which allowed me to share data across components without having to pass props at every level, thus making the code more maintainable and scalable. Another key part of the session was setting up React Router for client-side routing, enabling smooth navigation between different pages without reloading the entire application. Additionally, I used Axios to fetch data from an external API, which allowed me to integrate real-time data into my app. Overall, this session gave me a solid understanding of core React concepts like state management, data sharing, routing, and API integration, which are essential for developing interactive and modern web applications.